

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	03-10-2026	Duration:	60 minutes

Code of Conduct

I Tharika S / 192521140 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student Tharika S

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

Problem Statement: Predictive Analytics System for Early Detection of Land Acquisition Delays

Large public projects like highways, metro lines and railway corridors often get stuck because the land needed for them is not handed over on time. Notices get delayed, surveys are incomplete, owners file objections, court cases come up and compensation gets stuck in paperwork. Most of the time, the project office realises the delay only after the schedule is already lost. This document is the sprint planning and user story workshop for a system that studies land acquisition data, predicts which parcels are likely to be delayed, and warns the officers early so they can act. Scrum has been followed for the planning, with a team of five developers working in two-week sprints.

1. Identify the Project Vision and Stakeholders

Project Vision

Our vision is to give land acquisition officers and project managers a simple web based system that can look at the present status and past history of every land parcel and tell them, well in advance, which parcels are likely to get delayed. With this early warning, the officers can fix the problem (pending notice, missing document, unresolved objection) before it blocks the whole project.

Project Objectives

- Bring scattered acquisition data (Excel sheets, registers, reports) into one central system.
- Track every parcel through the acquisition stages from notification to possession.
- Calculate a delay risk score for each parcel, first with simple rules and later with a machine learning model.
- Send alerts when a parcel crosses the risk threshold so action can be taken at least 30 days before the planned date.
- Show dashboards and reports that help senior officers take decisions quickly.

Stakeholders

The people who are affected by, or who use, this system are listed below along with their role and what they expect from it.

Stakeholder	Role in the project	Main expectation
Project Director / PMO	Sponsor; owns the overall infrastructure project schedule	Early visibility of delay risks across all packages
Land Acquisition Officer (LAO)	Primary user; updates parcel stages and handles notices	Easy data entry and a clear list of risky parcels

District Revenue Authority	Provides land records and approves awards	Correct and traceable records
Survey and Field Team	Collects survey and measurement data	Simple way to submit survey status
Legal Team	Handles objections, court cases and stay orders	Case details linked to the parcel
Finance Department	Releases compensation to landowners	Advance notice of payment load
Landowners / Affected Families	Indirect stakeholders; affected by the acquisition	Fair and timely process (no direct login in this phase)
Contractors	Need land handed over to start construction	Reliable possession dates
Product Owner	Decides backlog priority from business side	Features that give maximum value first
Scrum Master	Facilitates Scrum events and removes blockers	Smooth sprint with no impediments
Development Team (5 members)	Design, build and test the system	Clear stories and realistic sprint scope
Data Analyst / ML Engineer	Prepares data and builds prediction model	Good quality historical data

Table 1: Stakeholders and their roles

2. Create Epics and User Stories

After discussing the problem with the team, we grouped the requirements into six epics. Each epic is a big feature area and is then broken into user stories written in the format: As a <user>, I want <feature>, so that <benefit>.

Epic ID	Epic Name	Description
E1	Data Integration and Management	Bringing parcel and historical acquisition data into the system and keeping it clean.
E2	Land Parcel and Acquisition Tracking	Tracking each parcel through the acquisition stages along with cases and disputes.
E3	Delay Prediction Engine	Calculating delay risk using rules first and then a machine learning model.
E4	Early Warning and Alerts	Notifying the right officers when a parcel becomes high risk.
E5	Dashboard and Reporting	Visual summary, maps and reports for decision makers.

E6	User Access and Security	Login, roles, permissions and audit of user actions.
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Table 2: Epics

User Stories

ID	Epic	User Story
US-01	E6	As a system administrator, I want to log in and give users role based access (Admin, LAO, Viewer), so that only authorised people can see or change land data.
US-02	E1	As a Land Acquisition Officer, I want to upload parcel data from an Excel/CSV file, so that I do not have to type hundreds of records manually.
US-03	E1	As a data analyst, I want the system to validate the uploaded data and show errors, so that wrong or duplicate records do not spoil the predictions.
US-04	E2	As a Land Acquisition Officer, I want to update the acquisition stage of a parcel, so that the current status is always correct and the history is kept.
US-05	E2	As a Land Acquisition Officer, I want to open a parcel and see its details and stage history, so that I can understand why it is stuck.
US-06	E3	As a Project Manager, I want a delay risk score (Low / Medium / High) for every parcel based on simple rules, so that I get a first idea of the risky parcels even before the ML model is ready.
US-07	E5	As a Project Director, I want a dashboard showing the number of parcels by risk level and stage, so that I can see the overall health of the project at a glance.
US-08	E2	As a legal officer, I want to record court cases and objections against a parcel, so that legal reasons for delay are available to the system.
US-09	E4	As a Land Acquisition Officer, I want an email alert when a parcel becomes high risk, so that I can act before the delay actually happens.
US-10	E3	As a data analyst, I want to train a machine learning model on historical acquisition data, so that predictions are based on real past patterns.
US-11	E3	As a Project Manager, I want to see the delay probability and the top reasons behind it, so that I know what to fix first.
US-12	E3	As a Project Director, I want to compare predicted delays with actual delays, so that I can trust (or question) the model's accuracy.
US-13	E5	As a Project Director, I want to download a monthly report in PDF/Excel, so that I can share it in review meetings.
US-14	E6	As an administrator, I want an audit log of who changed what, so that data changes can be traced during audits.
US-15	E5	As a Project Manager, I want to see parcels on a map coloured by risk, so that I can find problem areas along the project route.

Table 3: User stories mapped to epics

3. Define Acceptance Criteria

Acceptance criteria are written for the stories that we have selected for Sprint 1 (US-01, US-02, US-03, US-04, US-06 and US-07). They are in Given–When–Then format and each one has something we can actually measure or test, so there is no confusion at the time of the sprint review.

US-01: Login and role based access

Valid login: Given a registered user with a correct username and password, when the user clicks Login, then the home page opens within 3 seconds according to the user's role.

Invalid login: Given a wrong password, when the user tries to log in, then an error message is shown and the account is locked after 5 failed attempts.

Role restriction: Given a user with the Viewer role, when the user tries to edit a parcel, then the edit option is hidden and the server returns an access denied response.

US-02: Upload parcel data

Successful upload: Given a valid Excel/CSV file in the standard template, when the LAO uploads it, then all rows are saved and a message shows the count of records imported.

Large file: Given a file with 5,000 rows, when it is uploaded, then the import completes in less than 60 seconds.

Wrong format: Given a file that is not .xlsx or .csv, when it is uploaded, then the system rejects it and shows the allowed formats.

US-03: Validate imported data

Missing fields: Given a row with a missing survey number or village name, when the file is validated, then that row is marked as an error with the row number and reason.

Duplicates: Given two rows with the same survey number, when validation runs, then the duplicate is flagged and not imported.

Error report: Given a file with errors, when validation finishes, then the user can download an error list and the valid rows are still saved.

US-04: Update acquisition stage

Stage update: Given an existing parcel, when the LAO selects a new stage and saves, then the stage, date and user name are stored and the status shows on the parcel page immediately.

Order of stages: Given a parcel at the Notification stage, when the user tries to jump directly to Possession, then the system blocks it and shows the allowed next stage.

History: Given a parcel whose stage was updated three times, when the history is opened, then all three changes are listed with date and user.

US-06: Rule based delay risk score

High risk: Given a parcel whose stage has stayed unchanged for more than 60 days and which has a pending objection, when the score is calculated, then the risk is shown as High.

Low risk: Given a parcel moving on schedule with no objection or case, when the score is calculated, then the risk is shown as Low.

Recalculation: Given any change in parcel data, when the data is saved, then the risk score is recalculated and updated within 5 seconds.

US-07: Risk dashboard

Summary cards: Given parcels with scores available, when the Project Director opens the dashboard, then total parcels and counts of Low, Medium and High risk parcels are displayed.

Filtering: Given the dashboard is open, when the user filters by village or stage, then the table and counts update in less than 3 seconds.

Accuracy of numbers: Given 100 test parcels with known risk values, when the dashboard loads, then the numbers match the database exactly.

For the stories that are not in Sprint 1, the acceptance criteria will be written in detail during backlog refinement, just before they are picked for a sprint.

4. Prioritize the Product Backlog

The backlog was ordered by thinking about three things: the business value to the project office, what the customer (the LAO and Project Director) needs first, and the dependencies between stories. For example, the prediction features cannot be built until the data import and parcel tracking are ready, so those stories have to come first, even though prediction is the main goal of the project. The MoSCoW method (Must, Should, Could) was also used.

Rank	ID	Story (short)	MoSCoW	Reason for the position
1	US-01	Login and roles	Must	Needed by every other feature; security first
2	US-02	Upload parcel data	Must	Without data nothing else can work
3	US-03	Validate data	Must	Bad data gives bad predictions
4	US-04	Update acquisition stage	Must	Core tracking; stage is a main input for risk
5	US-06	Rule based risk score	Must	Gives early value to the client without waiting for ML

6	US-07	Risk dashboard	Must	Makes the score visible to decision makers
7	US-05	Parcel details and history	Should	Depends on US-04
8	US-08	Court cases and objections	Should	Strong cause of delay; improves the risk score
9	US-09	Email alerts	Should	Depends on risk score (US-06)
10	US-10	Train ML model	Must	Needs enough clean data from earlier sprints
11	US-11	Probability and reasons	Should	Depends on US-10
12	US-12	Predicted vs actual	Should	Needs some completed parcels and the ML model
13	US-13	Monthly report	Could	Useful but dashboard already gives the main view
14	US-14	Audit log	Could	Good for audit, not blocking other features
15	US-15	Map view	Could	High effort and needs GIS data; planned for later

Table 4: Prioritized product backlog

5. Plan the Sprint Backlog

Sprint 1 is planned for two weeks (10 working days). The team has 5 members and each one can give about 6 productive hours per day, which gives roughly 300 hours. After keeping time for the Scrum meetings and some buffer, around 240 hours are planned for development work. The top six stories from the backlog (rank 1 to 6) were picked as they are the highest priority and together fit the expected velocity of about 30 story points.

Task breakdown

Story	Task	Owner (role)	Hours
US-01	Design users and roles tables	Backend	4
	Build login API with password hashing and token	Backend	8
	Create login page	Frontend	5
	Test role permissions	Tester	3
US-02	Decide the Excel/CSV template columns with LAO	Analyst	3

	Build upload API and file parser	Backend	10
	Create upload screen with progress message	Frontend	6
	Test with a sample file of 500 and 5,000 rows	Tester	4
US-03	List the validation rules	Analyst	3
	Code checks for missing fields, duplicates and dates	Backend	8
	Create downloadable error report	Backend	5
	Test with deliberately wrong files	Tester	3
US-04	Create stage master and parcel stage history tables	Backend	5
	Build stage update API with order check	Backend	8
	Create stage update form	Frontend	7
	Test stage rules	Tester	4
US-06	Identify risk factors with the LAO	Analyst	6
	Decide rules and weights for Low / Medium / High	Analyst	6
	Code the scoring function	Backend	12
	Save and recalculate scores on data change	Backend	4
	Unit tests using sample parcel cases	Tester	8
	Review rules with domain expert	Analyst	4
US-07	Draw wireframe of dashboard	Frontend	4
	Build summary API (counts by risk and stage)	Backend	8
	Create dashboard cards and parcel table	Frontend	12
	Add filters by village and stage	Frontend	6
	Test numbers against database	Tester	4
Setup	Project setup, database, Git repository	Team	10
	Bug fixing, sprint demo preparation	Team	14

Table 5: Sprint 1 tasks (total planned effort is about 184 hours)

The total task effort comes to about 184 hours, which is less than the 240 hours available. We kept this gap on purpose for unplanned work, code reviews and any clarification needed from the officers, because requirements around land records are usually not very clear in the beginning.

6. Estimate Effort Using Story Points

The Fibonacci sequence (1, 2, 3, 5, 8, 13) was used for estimation through Planning Poker. Each team member picked a card privately, and when the cards were very different (for example 3 and 8), the people with the highest and lowest numbers explained their thinking and the round was repeated till the team came close to one value. US-01 (login) was taken as the

reference story with 3 points because everyone had built something similar before, and the other stories were compared with it.

ID	Story	Points	Justification
US-01	Login and role access	3	Standard feature; reference story. Small amount of uncertainty.
US-02	Upload parcel data	5	File parsing and handling different file formats and sizes need extra effort.
US-03	Validate imported data	3	Rule checks are simple, but needs a good error report.
US-04	Update acquisition stage	5	Needs stage order rules and history, so more logic than basic CRUD.
US-06	Rule based risk score	8	Highest uncertainty; rules must come from domain experts and need repeated review.
US-07	Risk dashboard	5	UI work with filters and API; depends on scores being ready.
	Total for Sprint 1	29	Close to the expected velocity of 30 points

Table 6: Story point estimates for Sprint 1

The remaining backlog items were also given rough estimates: US-05 = 3, US-08 = 5, US-09 = 5, US-10 = 13, US-11 = 8, US-12 = 5, US-13 = 5, US-14 = 3 and US-15 = 8. US-10 got the biggest value (13) because model training, data preparation and testing are all unknown at this point, and it will be split into smaller stories before being taken in a sprint. These estimates will be adjusted after the team sees its actual velocity in Sprint 1.

7. Present the Sprint Goal and Expected Deliverables

Sprint Goal

By the end of Sprint 1, a logged-in officer should be able to upload and validate land parcel data, update the acquisition stage of each parcel, and view an automatically calculated rule based delay risk (Low, Medium or High) for every parcel on a dashboard.

Expected Deliverables

At the end of the two weeks, the following working features will be shown to the Product Owner and stakeholders in the Sprint Review:

- Secure login with Admin, LAO and Viewer roles (US-01).
- Excel/CSV upload of parcel data with a downloadable error report (US-02, US-03).
- Stage update screen with proper stage order and stored history (US-04).
- Rule based risk score calculated for each parcel (US-06).

- Dashboard with risk counts, stage-wise table and filters (US-07).
- Test cases, a short user guide for the upload template and the source code in the Git repository.

Definition of Done

A story will be treated as done only when the code is reviewed and merged, all acceptance criteria are tested and passed, no high priority bugs are open, and the Product Owner has accepted it in the sprint review.

Risks and Assumptions

The main risk is that historical land records may be incomplete or in different formats, which can slow down US-02 and US-03. To reduce this, a sample file will be collected from the LAO in the first two days. We have assumed that a domain expert will be available for rule discussion in the first week, and that no team member will take long leave during the sprint. The ML part is planned for Sprint 3, once enough clean data is stored.

At the end of the sprint we will hold the Sprint Review and a Retrospective to check what went well, what did not, and how to improve in Sprint 2, where US-05, US-08 and US-09 are the likely starting stories.